

Yashpal Singh

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PROFESSIONAL SUMMARY

AI/ML R&D Engineer with 4 years of professional experience building and deploying intelligent applications and AI-driven solutions. Experienced in Python, Machine Learning, LLMs, RAG, FastAPI, OpenCV, and modern AI frameworks. Skilled in developing practical AI systems, integrating models into production applications, and solving real-world problems through applied AI/ML.

EXPERIENCE

AI/ML Developer (Projects & Research)

Aug 2022 - Present

Phibonacci learning | Ahmedabad

- Designed and developed AI/ML solutions using Python, machine learning algorithms, and modern AI frameworks.
- Developed RAG-based applications using LLM workflows, vector databases, embeddings, and retrieval pipelines.
- Worked on model training, fine-tuning, and inference workflows using PyTorch and Hugging Face.
- Developed computer vision solutions using OpenCV and YOLO-based models for real-world applications.
- Built and integrated AI/ML APIs and services using FastAPI for application and production workflows.

EDUCATION

Polytechnic - Computer Science and Engineering

Aug 2019 - Aug 2022

M G Polytechnic (Hathras)

B.Tech - Computer Science and Data Science

Aug 2022 - Aug 2025

Dr. A.P.J. Abdul Kalam Technical University

- Bachelor of Technology in Computer Science and Data Science with focus on Artificial Intelligence, Machine Learning, and real-world application development. Developed academic and self-driven projects to strengthen skills in problem solving, model development, and intelligent systems.

SKILLS

Python | Machine Learning | Deep Learning | Model Training | Fine-Tuning | LLMs | LangChain | RAG | Prompt Engineering | PyTorch | TensorFlow | Hugging Face | OpenCV | FastAPI | YOLO | Vector Databases | Git | GitHub | NumPy | Pandas | Natural Language Processing (NLP) | Model Evaluation | Generative AI | Feature Engineering | Embeddings | Computer Vision

PROJECTS

CVKing - AI Resume & Career Platform

AI/ML Developer & Full-Stack Developer

<https://cvking.in/>

- Independently built and deployed a production-ready AI-powered resume and career platform from scratch.
- Developed AI-driven resume analysis, ATS evaluation, and intelligent resume generation features.
- Integrated LLM-based workflows and AI services with backend APIs, database, and application workflows.
- Designed and implemented the complete frontend, backend, database architecture, and deployment.

Billo AI – RAG-Based AI Assistant

AI/ML Developer

- Built a Retrieval-Augmented Generation (RAG) pipeline for contextual question answering over user-provided documents.
- Implemented document processing, embeddings, semantic retrieval, and vector search using LangChain and ChromaDB.
- Integrated LLM workflows with FastAPI to build an AI assistant capable of generating context-aware responses.
- Developed the system with a focus on modular AI pipelines, retrieval quality, and practical deployment.

Computer Vision System

AI/ML Developer

- Developed computer vision solutions using Python, OpenCV, and YOLO-based object detection models.
- Worked on image processing, model inference, and integrating computer vision capabilities into applications.

Machine Learning / NLP Project

Machine Learning Developer

- Developed a machine learning-based NLP solution for detecting fraudulent/spam SMS messages.
- Applied text preprocessing, feature extraction, model training, and evaluation to classify messages.